

Exercise 4 – OpenAI API in Python (Jupyter Notebook)

This exercise uses the OpenAI Chat Completions API (gpt-3.5-turbo) in a Jupyter notebook to explore system and user prompts. It includes the notebook setup, the given example prompt pairs, and a custom prompt pair.

1. Setup

L1 Language Models, the Chat Format and Tokens

Setup

Load the API key and relevant Python libraries.

In this course, we've provided some code that loads the OpenAI API key for you.

```
In [6]: import os
import openai
import tiktoken
from dotenv import load_dotenv, find_dotenv
_ = load_dotenv(find_dotenv()) # read local .env file

openai.api_key = os.environ['OPENAI_API_KEY']
```

helper function

This may look familiar if you took the earlier course "ChatGPT Prompt Engineering for Developers" Course.

Throughout this course, we will use OpenAI's `gpt-3.5-turbo` model and the [chat completions endpoint](#).

This helper function will make it easier to use prompts and look at the generated outputs.

Note: In June 2023, OpenAI updated gpt-3.5-turbo. The results you see in the notebook may be slightly different than those in the video. Some of the prompts have also been slightly modified to produce the desired results.

Figure 1: Notebook setup — importing libraries and loading the OpenAI API key.

2. Prompt Pair 1 — Physics Assistant (Given Example)

System prompt: "You are a helpful assistant with deep knowledge in physics."

User prompt: "Explain how black holes are formed."

```
In [3]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a helpful assistant with deep knowledge in physics."}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "Explain how black holes are formed."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Black holes are formed when a massive star runs out of fuel and collapses in on itself during a supernova explosion. As the star collapses, the outer layers are blasted into space while the core collapses under its own gravity. If the core has a mass greater than about three times that of the Sun, it will collapse into a black hole.

The core of the star becomes so dense that its gravity becomes incredibly strong, causing it to collapse to a point of infinite density called a singularity. This singularity is surrounded by an event horizon, which is the point of no return for anything that crosses it. Once an object or light crosses the event horizon, it can never escape the gravitational pull of the black hole.

Black holes come in different sizes, with stellar-mass black holes being formed from the collapse of massive stars, and supermassive black holes found at the centers of galaxies, with masses millions or even billions of times that of the Sun.

Figure 2: Response explaining how black holes form, using the physics-assistant system prompt.

3. Prompt Pair 2 — Event Planner (Given Example)

System prompt: "You are a expert event planner."

User prompt: "How to plane my friend's birthday party."

```
In [4]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a expert event planner."}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "How to plane my friend's birthday party."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Planning a birthday party for your friend can be a lot of fun! Here are some steps to help you organize a memorable celebration:

1. ****Set a Budget**:** Determine how much you are willing to spend on the party, and allocate funds for venue, decorations, food, entertainment, and other expenses.
2. ****Choose a Date and Time**:** Consult with your friend to pick a date and time that works best for them and their guests.
3. ****Select a Venue**:** Decide whether the party will be held at a home, restaurant, park, or any other location that suits the theme and guest count.
4. ****Theme and Decorations**:** Choose a theme that reflects your friend's interests and personality. Decorate the venue with balloons, banners, tableware, and other decorations that fit the theme.
5. ****Invitations**:** Send out invitations to the guests well in advance. You can choose to send physical invitations, digital invites, or create a Facebook event.
6. ****Food and Drinks**:** Plan the menu based on your friend's preferences. You can opt for catering, potluck style, or cook the food yourself. Don't forget the birthday cake!
7. ****Entertainment**:** Organize activities or games that your friend and the guests will enjoy. You can also hire a DJ, plan a movie marathon, or set up a DIY photo booth for some fun.
8. ****Gifts and Party Favors**:** Encourage guests to bring gifts for the birthday celebrant. You can also prepare small party favors as a token of appreciation for the attendees.
9. ****Timeline**:** Create a timeline for the party to ensure that everything runs smoothly. Include time for food service, games, cake cutting, and gift opening.
10. ****Capture Memories**:** Don't forget to document the party with photos and videos. You can hire a photographer or ask a friend to take pictures throughout the event.

Remember, the most important thing is to make your friend feel special and loved on their birthday. Have fun planning the party, and your friend will surely appreciate the effort you put into making their day memorable!

Figure 3: Response giving a 10-step birthday party planning guide.

4. Prompt Pair 3 — Custom: Budget Travel Agent

System prompt: "You are a friendly travel agent who specializes in budget backpacking trips."

User prompt: "I have a 5-day trip to Sri Lanka and a budget of \$150. Suggest an itinerary."

```
In [5]: # Define system prompt (initial behavior)
system_prompt = {"role": "system", "content": "You are a friendly travel agent who specializes in budget backpacking trips"}

# Define user prompt (specific query)
user_prompt = {"role": "user", "content": "I have a 5-day trip to Sri Lanka and a budget of $150. Suggest an itinerary."}

# Send prompt to API
response = openai.ChatCompletion.create(
    model="gpt-3.5-turbo",
    messages=[system_prompt, user_prompt]
)
print(response['choices'][0]['message']['content'])
```

Sure, here's a suggested itinerary for your 5-day trip to Sri Lanka on a \$150 budget:

Day 1:

- Arrival in Colombo
- Explore the city on foot or by local transportation
- Visit Galle Face Green, Viharamahadevi Park, and Gangaramaya Temple
- Stay in a budget hostel or guesthouse in Colombo

Day 2:

- Take a train or bus to Kandy
- Visit the Temple of the Sacred Tooth Relic and wander around the Kandy Lake
- Explore the local markets and try some street food
- Stay in a budget accommodation in Kandy

Day 3:

- Visit the famous Botanical Gardens in Peradeniya
- Take a walk around town and visit the Udawattakele Sanctuary
- Enjoy a traditional Sri Lankan meal at a local eatery
- Stay in Kandy

Day 4:

- Travel to Nuwara Eliya, known as "Little England"
- Visit a tea plantation and learn about the tea-making process
- Explore the scenic landscapes and waterfalls in the area
- Stay in a budget guesthouse in Nuwara Eliya

Day 5:

- Take a scenic train ride from Nuwara Eliya to Ella
- Explore Ella's famous attractions such as Little Adam's Peak and Nine Arch Bridge
- Enjoy the stunning views and relax in the laid-back atmosphere of Ella
- Return to Colombo for your departure

Please note that this itinerary is based on a tight budget, so make sure to allocate your funds wisely for transportation, accommodation, meals, and entrance fees to attractions. Additionally, consider using public transportation and eating at local eateries to save money. Enjoy your trip to Sri Lanka!

Figure 4: Response giving a 5-day budget itinerary for Sri Lanka.